LAB4 – Web Session Analysis (DNS, TCP and HTTP)

Course: The Bits and Bytes of Computer Networking
Author: Florencia Horita
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1. Introduction

In computer networks, accessing a web service involves the interaction of multiple protocols operating at different layers of the network model.

This laboratory analyzes the complete process that occurs when a client accesses a web server within a local network. The *Simulation Mode* of Cisco Packet Tracer is used in order to observe the packet exchange between network devices. During the simulation, the following protocols are analyzed:

- **Address Resolution Protocol (ARP)**
- **Domain Name System (DNS)**
- **Transmission Control Protocol (TCP)**
- **Hypertext Transfer Protocol (HTTP)**

The objective of this lab is to understand how these protocols interact to allow a client to access a web page using a domain name.

2. Network Topology

The network used in this laboratory consists of three devices connected within the same local network:

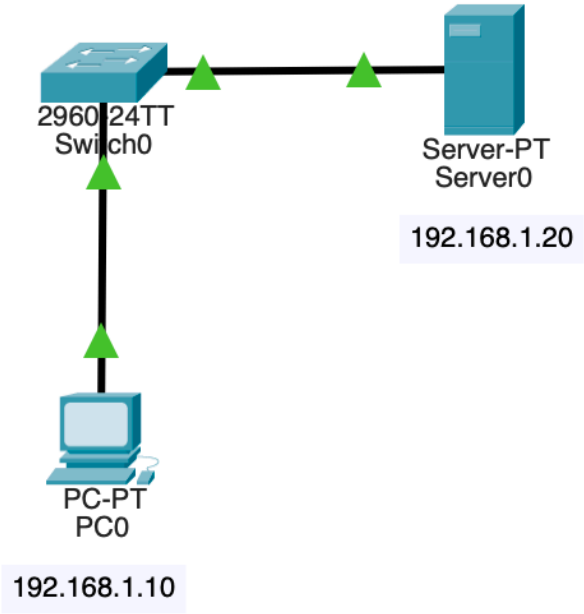
- 1 PC (client)
- 1 Cisco 2960 switch
- 1 server

Physical connections:

PC0 Fa0 → Switch0 Fa0/1
Switch0 Fa0/2 → Server0 Fa0

The client accesses the web server through the switch within the local network.

Figure 1 – Network topology used in the laboratory



3. Network Configuration

3.1 IP Addressing

The server performs two functions:

• DNS server

• HTTP server

Device	IP Address	Subnet Mask	DNS
PC0	192.168.1.10	255.255.255.0	192.168.1.20
Server0	192.168.1.20	255.255.255.0	—

3.2 DNS Service Configuration

The DNS service was enabled on the server and an **A record** was created to associate the domain name with the server's IP address.

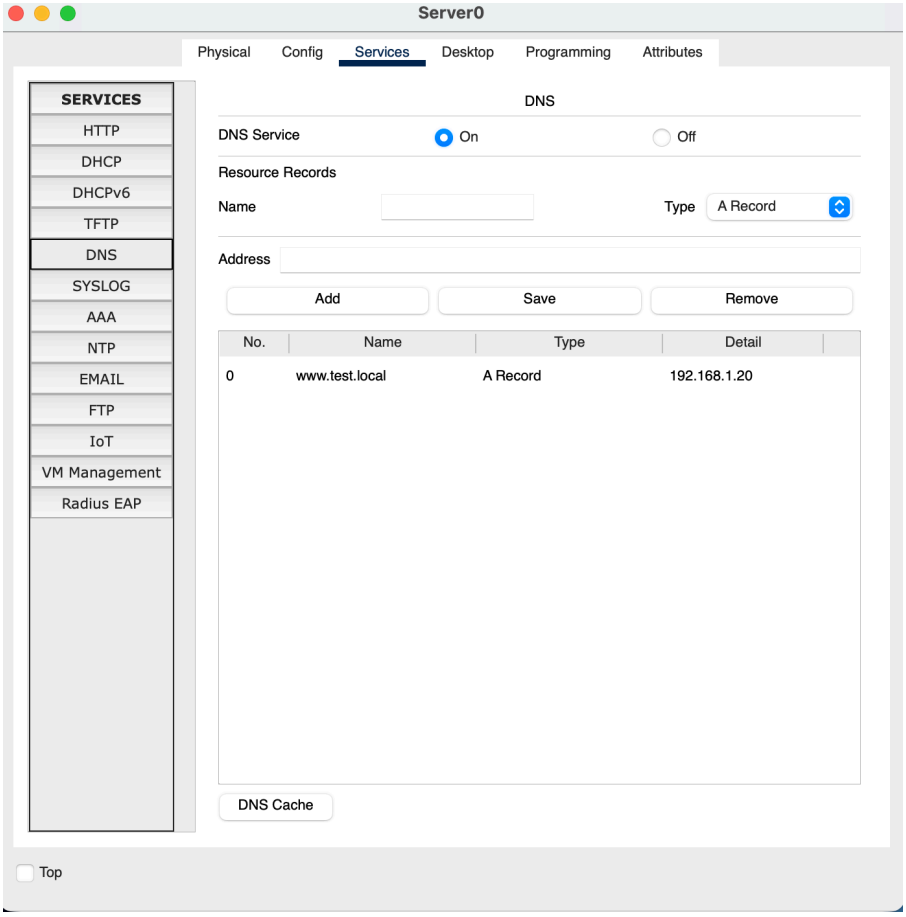
Name	Type	Address
www.test.local	A Record	192.168.1.20

In this way, when the client requests the domain:

www.test.local

the DNS server responds with the corresponding IP address.

Figure 2 – DNS service configuration



3.3 HTTP Service Configuration

The HTTP service was enabled on the server and the **index.html** file was modified to display the following content:

Web Server Running

Networking Lab – Protocol Analysis

PC Client → Switch → Server

This allows verification that the server responds correctly when the client accesses the web page.

4. Protocol Analysis in Simulation Mode

Using the Simulation Mode in Cisco Packet Tracer, the sequence of protocols involved when the client accesses the web server can be observed.

The observed sequence was:

ARP → DNS → TCP → HTTP

4.1 ARP Resolution

Initially, the client needs to know the MAC address of the server within the local network.

To achieve this, the Address Resolution Protocol (ARP) is used to resolve the MAC address corresponding to the server's IP address.

4.2 DNS Resolution

Once communication at the data link level is established, the client sends a DNS query to resolve the domain:

www.test.local

The DNS server responds with the IP address:

192.168.1.20

This allows the client to determine the address of the web server it wants to connect to.

4.3 TCP Connection Establishment

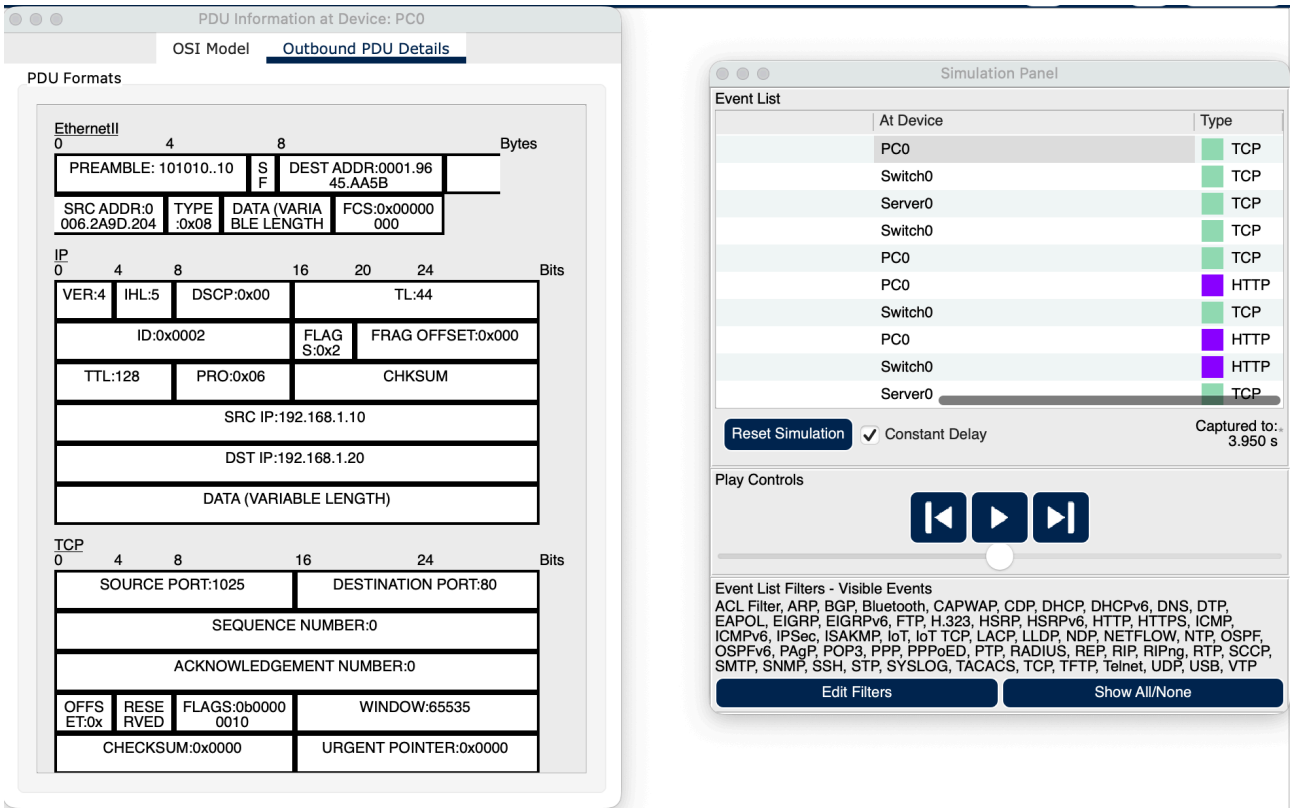
After obtaining the server's IP address, the client initiates a TCP connection with the web server.

This process is performed through the **Three Way Handshake**:

1. SYN – the client requests to initiate the connection
2. SYN-ACK – the server responds accepting the connection
3. ACK – the client confirms the connection establishment

As shown in **Figure 3**, the client sends the first TCP packet with the SYN flag, initiating the connection establishment process.

Figure 3 – TCP connection establishment (Three Way Handshake)



4.4 HTTP Request

Once the TCP connection is established, the client sends an HTTP request to the server to retrieve the requested resource.

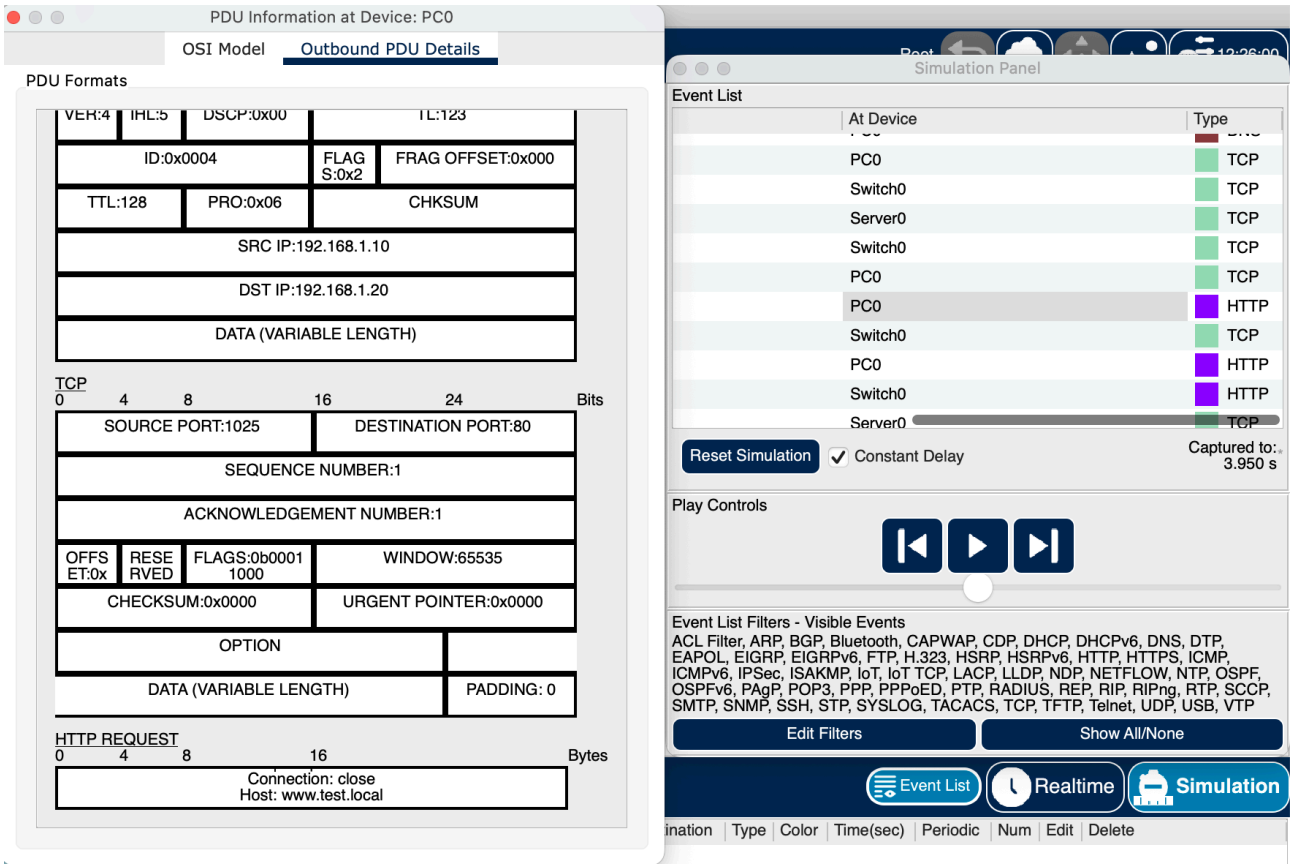
In the HTTP packet details, the following header can be observed:

Host: www.test.local

This header indicates to the server which resource is being requested.

As shown in **Figure 4**, the client sends the HTTP request to the server using TCP destination port 80.

Figure 4 – HTTP request sent from the client to the server



5. Web Service Verification

After the packet exchange is completed, the server responds with the configured web page.

The client accesses the server using the web browser and the address:

http://www.test.local

As shown in **Figure 5**, the web page loads successfully displaying the content configured on the server.

Figure 5 – Web page successfully served to the client

6. Results

During this laboratory, the operation of the protocols involved in accessing a web server within a local network was verified.

The complete communication flow between the client and the server was observed:

ARP → DNS → TCP → HTTP

The client successfully resolved the server name using DNS, established a TCP connection through the Three Way Handshake process, and finally sent an HTTP request to retrieve the web page.

The page configured on the server was successfully loaded in the client's browser.

7. Conclusion

This laboratory allowed the analysis of the complete communication process between a client and a web server within a local network.

The behavior of the DNS, TCP, and HTTP protocols was observed, as well as the way they interact to enable access to a web service using a domain name.

The use of the Simulation Mode in Cisco Packet Tracer made it possible to inspect the packets exchanged between network devices and understand the establishment of a TCP connection through the Three Way Handshake, as well as the subsequent HTTP request sent by the client.

This type of analysis helps to understand how transport and application layer protocols work together to provide network services.

Included Files

- LAB4_WEB_SESSION_ANALYSIS_FLOR.pkt – Cisco Packet Tracer project
- LAB4_WEB_SESSION_ANALYSIS_FLOR.pdf – laboratory documentation